Interstellar Harvest: Growing Food Beyond Earth

COURSE SYLLABUS

Format: Self-paced, fully online
Platform: TalentLMS
Modules: 3
Est. Time: 3 to 4 hours

1 Course Title and Description

COURSE Interstellar Harvest: Growing Food Beyond Earth	FORMAT Self-paced, fully online
PLATFORM TalentLMS	3 MODULES TO COMPLETE Est. 3 to 4 hours

Interstellar Harvest is a simulation-based minicourse exploring the science of space agriculture. Through three interactive modules, learners explore why Earth agriculture fails in space, engineer hydroponic systems designed to work around those constraints, and manage a lunar greenhouse through a full crop growth cycle. Each module is built around an interactive simulation that puts learners in the role of a space farmer making real decisions under mission constraints.

The course is designed for curious learners of all ages. No science background is required. Learners who complete all three modules will have a working understanding of the biological, engineering, and operational challenges of growing food beyond Earth. They will also understand why solving those challenges matters for the future of long-duration spaceflight.

2 Course Learning Outcomes

By the end of this course, learners will be able to:

- CLO 1 Describe the basic challenges of growing plants in space
- CLO 2 Identify at least three plant types that could grow well in space
- CLO 3 Recognize common problems in plant growth and basic solutions
- CLO 4 Explain how growing plants in space could help astronauts on long-duration missions
- CLO 5 Explain new farming technologies and how they could be used in space
- CLO 6 Suggest simple ways to grow more plants with limited resources

3 Course Alignment Map

This table shows how each course learning outcome is supported by module activities and assessed throughout the course.

LEARNING OBJECTIVE	ACTIVITY	ASSESSMENT
Describe the basic challenges of growing plants in space (CLO 1)	Read: The Challenge of Growing Food in Space. Use the Destination Comparator simulation to compare crop outcomes across Earth, Moon, Mars, and ISS.	Module 1 Knowledge Check, Mission Briefing assignment
Explain how growing plants in space could help astronauts on long-duration missions (CLO 4)	Watch: Why Do We Grow Plants in Space? Participate in Discussion 2: Space Farming and the Long Game.	Mission Briefing assignment, Discussion 2 scored with discussion rubric
Identify at least three plant types that could grow well in space (CLO 2)	Read: Engineering the Solution. Use the Hydroponic Systems Lab simulation to design grow systems for different crops.	Module 2 Knowledge Check, System Design Brief assignment
Recognize common problems in plant growth and basic solutions (CLO 3)	Read: Engineering the Solution. Use the Hydroponic Systems Lab simulation to diagnose and fix system failures across five scenarios.	Module 2 Knowledge Check, System Design Brief assignment
Explain new farming technologies and how they could be used in space (CLO 5)	Watch: Inside NASA's Kennedy Space Center. Use the Hydroponic Systems Lab and Lunar Greenhouse simulations. Participate in Discussion 2: Space Tech Comes Back to Earth.	Module 2 and Module 3 Knowledge Checks, System Design Brief, Mission Operations Report
Suggest simple ways to grow more plants with limited resources (CLO 6)	Read: Running the Farm. Use the Lunar Greenhouse simulation to manage a full crop growth cycle through six phase transitions. Participate in Discussion 2: The Full Picture.	Module 3 Knowledge Check, Mission Operations Report assignment

4 Instructor Contact

INSTRUCTOR
N. Weis

EMAIL
NWEIS@umgc.edu

RESPONSE TIME
48 hours for discussions, 2 to 3 business days for email

Use the discussion forums inside each module for content questions. This keeps the conversation visible to all learners and often means a peer will respond before I do. For personal or technical issues, email is preferred.

5 Course Prerequisites

There are no formal prerequisites for this course. Interstellar Harvest is designed for curious learners of all ages and backgrounds. No prior knowledge of science, biology, agriculture, or space exploration is required.

Learners should be comfortable navigating a web browser and have access to a reliable internet connection. The simulations in this course run directly in the browser on desktop and laptop computers. Chrome and Firefox work best. Mobile devices can access most course content but the simulations are optimized for larger screens.

6 Technology Requirements

Device

A desktop or laptop computer is strongly recommended. The course simulations are designed for larger screens and may not display correctly on phones or tablets. Most reading content, videos, and knowledge checks can be completed on a mobile device.

Browser

Google Chrome or Mozilla Firefox, updated to the latest version. Safari and Edge are generally compatible but Chrome and Firefox are recommended for the best simulation experience.

Internet Connection

A stable broadband connection is required. The simulations and embedded videos require sufficient bandwidth to load without interruption.

Software

No downloads, plugins, or additional software are required. All course content runs directly in the browser.

PDF Viewer

The course syllabus and assessment rubrics are available as PDF files. Most browsers open PDFs natively. If yours does not, a free PDF viewer such as Adobe Acrobat Reader is sufficient.

Simulation Sessions

The course simulations do not save progress between sessions. Plan to complete each simulation in a single sitting.

TalentLMS Account

A free TalentLMS account is required to access the course. Visit interstellarharvest.talentlms.com and click Sign Up to register. You can sign up with an email address or log in directly with a Google account.

Accessibility

All videos in this course include closed captions. Reading content is compatible with screen readers. If you encounter any accessibility barriers, contact the instructor at NWEIS@umgc.edu.

7 Grading Policy

Graded Elements

ELEMENT	POINTS EACH	MODULES	TOTAL
Knowledge Check	10 pts	3	30 pts
Discussion 1	15 pts	3	45 pts
Discussion 2	15 pts	3	45 pts
Module Assignment	25 pts	3	75 pts
End-of-Course Survey	5 pts	1	5 pts
Course Total			**200 pts**

Grade Scale

GRADE	PERCENTAGE	POINTS
A	90–100%	180–200 pts
B	80–89%	160–179 pts
C	70–79%	140–159 pts
Below C	Below 70%	Below 140 pts

Notes on Grading

Knowledge checks have unlimited attempts. Your highest score is recorded. There is no penalty for retaking a knowledge check.

Discussions are scored using the Universal Discussion Rubric. Both your initial post and your reply to a peer are required for full credit. An initial post without a peer reply will receive a maximum of 80% of available points.

Module assignments are scored using the module-specific assignment rubric. Rubrics are linked directly above each assignment inside the course.

The End-of-Course Survey is scored on completion only. Any submitted response receives full credit.

8 Late Policies

This course is self-paced with no fixed due dates for individual activities. Learners are encouraged to work through each module in order and complete all activities before moving to the next module, but there is no penalty for taking additional time.

Incomplete Work

All graded activities must be submitted to receive a final course score. Knowledge checks, discussions, and assignments that are skipped will receive a score of zero and will lower your overall course grade accordingly.

Resubmission

Knowledge checks may be retaken unlimited times. Discussion posts and module assignments may not be resubmitted after initial submission. Review the rubric before submitting to make sure your work meets the expectations for the grade you are aiming for.

Simulation Runs

The simulations do not save progress if you close the browser, refresh the page, or log out of TalentLMS. Plan each simulation session accordingly. If a technical issue causes you to lose progress mid-simulation, contact the instructor at NWEIS@umgc.edu before attempting to complete the downstream discussion or assignment so we can account for the disruption.

Extensions

Because this course is self-paced, formal extension requests are not necessary. If you encounter a technical problem, accessibility barrier, or personal circumstance that prevents you from completing the course, contact the instructor at NWEIS@umgc.edu and we will work out a solution.

9 Schedule of Instructional Events

This course is self-paced with no fixed due dates. The schedule below shows the recommended order of completion for each module and its activities. Work through each module in order before moving to the next.

MODULE	ACTIVITY	TYPE
Module 1: Why Space Farming?	Introduction	Reading
	Module Opening Screencast	Video
	The Challenge of Growing Food in Space	Reading
	Why Do We Grow Plants in Space?	Video
	Before You Begin: Destination Comparator	Reading
	Destination Comparator Simulation	Interactive
	Discussion 1: What Did Your Crops Tell You?	Discussion
	Discussion 2: Space Farming and the Long Game	Discussion
	Module 1 Knowledge Check	Assessment
	Mission Briefing Assignment	Assignment
Module 2: Building the System	Introduction	Reading
	Engineering the Solution	Reading
	Growing Food in Space: Artemis, Mars and Beyond	Video
	Before You Begin: Hydroponic Systems Lab	Reading
	Hydroponic Systems Lab Simulation	Interactive
	Discussion 1: What Did You Build and Why?	Discussion
	Discussion 2: Space Tech Comes Back to Earth	Discussion
	Module 2 Knowledge Check	Assessment
	System Design Brief Assignment	Assignment
Module 3: Running the Farm	Introduction	Reading
	Running the Farm	Reading
	Growing Chile Peppers in Space	Video
	Before You Begin: Lunar Greenhouse	Reading
	Lunar Greenhouse Simulation	Interactive
	Discussion 1: What Did Your Greenhouse Tell You?	Discussion
	Discussion 2: The Full Picture	Discussion
	Module 3 Knowledge Check	Assessment
	Mission Operations Report Assignment	Assignment
Course Wrap-Up	End-of-Course Survey	Survey

10 Academic Honesty and Integrity Policy

Interstellar Harvest is a learning experience built around personal engagement with simulations, discussions, and reflective writing. The value of this course depends on learners doing their own thinking, not presenting someone else's work as their own.

Expectations

Discussion posts should reflect your own observations and reasoning. Reading what peers have written before posting your own initial response undermines the purpose of the discussion design, which asks you to commit to your own thinking before seeing others.

Module assignments ask you to synthesize your own experience with the simulations. Submitting work that was written by another person or generated entirely by an AI tool without meaningful personal contribution does not demonstrate the learning the assignment is designed to assess.

On the Use of AI Tools

AI tools may be used to help organize ideas, improve writing clarity, or explore background information. Any content generated with AI assistance should be revised to reflect your own thinking and clearly attributed. Submitting AI-generated text as your own original work without attribution or meaningful engagement is a violation of this policy.

On Collaboration

Discussing ideas with other learners is encouraged. Submitting identical or near-identical work as another learner is not. Your written submissions should be your own.

Consequences

Violations of this policy may result in a zero on the affected assignment and removal from the course at the instructor's discretion.

Questions

If you are unsure whether a particular use of a source or tool is appropriate, contact the instructor at NWEIS@umgc.edu before submitting.

11 Accommodations for Students with Disabilities

Interstellar Harvest is designed with accessibility in mind. The following features are built into the course to support diverse learners:

- All videos include closed captions
- All images include descriptive alt text
- Reading content is compatible with screen readers
- The course syllabus and rubrics are available as PDF files designed for screen reader accessibility
- Knowledge check feedback is provided in text form for all question types
- Color is not used as the sole means of conveying meaning in any course material
- All external links open in a new tab and are described in writing

If You Need Additional Accommodations

If you have a disability or accessibility need that is not addressed by the features listed above, contact the instructor at NWEIS@umgc.edu before beginning the course. We will work together to identify appropriate adjustments.

A Note on the Simulations

The course simulations are interactive browser-based tools that rely on mouse or trackpad input and visual feedback. Full keyboard navigation is not currently available in all simulations. If this presents a barrier, contact the instructor and an alternative pathway through the module content will be provided.

A Note on Captions

Video captions in this course are provided through YouTube's captioning system. If you find any captions to be inaccurate or incomplete, please notify the instructor so corrections can be requested.

12 Copyright, Licensing, and PDF Availability

Creative Commons License

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Third-Party Content

Videos embedded in this course are hosted on YouTube and are used in accordance with their respective licenses. All videos are attributed within the module where they appear. Reading content references are cited in APA format at the end of each module's reading unit.

PDF Version of This Syllabus

A PDF version of this syllabus is available for download here: Interstellar Harvest Syllabus (PDF)

> NOTE
> If the PDF link does not work or the file does not open, contact the instructor at NWEIS@umgc.edu and a copy will be sent directly.